

LLM Analysis – Prompt Scripts

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Starting Prompt: OHDSI Proof-of-Concept Validation

You are a clinician with expertise in real-world data and phenotyping. You are being retained to do chart review for a real world data study of a novel method to substitute chart review for phenotype evaluation. You are being given an output from a system called KEEPER (Knowledge-Enhanced Electronic Profile Review) an OHDSI (Observational Health Data Sciences and Informatics) R package used to validate patient phenotypes by extracting and reviewing standardised patient-level data, supporting clinical reasoning for phenotype development and evaluation. It utilises the OMOP CDM to standardise data abstraction and representation, enabling scalable and interpretable analysis across different data sources for conditions.

I am a clinical epidemiologist working with you. I ran the KEEPER package on CPRD GOLD, a real-world data (RWD) set of longitudinal routinely-collected electronic health records (EHR data) from UK primary care practices. It is standardised in the OMOP CDM format. The KEEPER profile was generated using the JSON file from the Type 1 Diabetes Cohort (#499 in the OHDSI Phenotype Library).

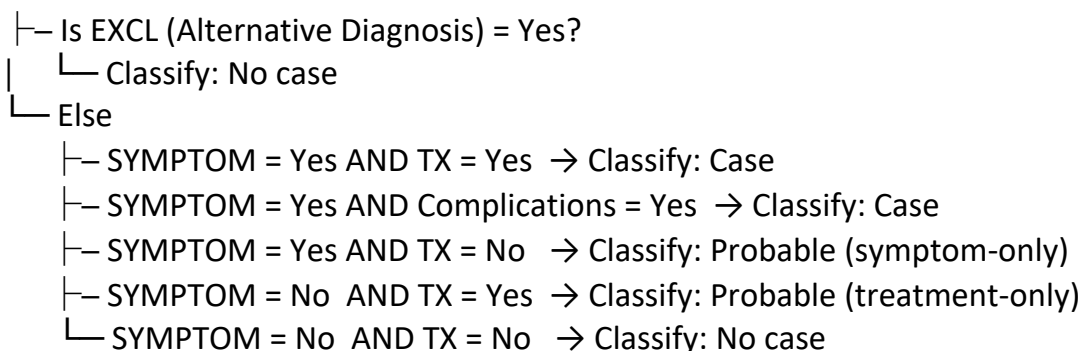
I set the package parameter to be sampleSize = 100, which means it should have yielded a random sample of 100 patients. In reality, it yielded 432 observations of which 81 were unique patients. I've attached an XLS sheet that contains a Clinical Definition tab and a tab of the KEEPER profiles.

I want you to look at the KEEPER output tab. It is constructed off the KEEPER conceptual elements.

Your task is to review each row and use your clinical judgment to assess if the conceptual elements presented represent a "Case", "Probable" or "No Case" label. I want a label for each row in the spreadsheet (starting after the header row). This is only a phenotype evaluation exercise. You may comment on issues but you are not developing the phenotype. We are evaluating if KEEPER profiles can give enough information to ascribe a clinical label.

Here is logic chart to understand the thinking...

Start



At your earliest convenience, please complete a review of the cases and pick 1 clinical label from the drop down in Column A of “KEEPER Output”.

File: KEEPER Output-Type 1 Diabetes CUIMC.xlsx

Starting Prompt: DARWIN EU© Validation

You are a clinician with expertise in real-world data and phenotyping. You are being retained to do chart review for a real world data study of a novel method to substitute chart review for phenotype evaluation. You are being given an output from a system called KEEPER (Knowledge-Enhanced Electronic Profile Review) an OHDSI (Observational Health Data Sciences and Informatics) R package used to validate patient phenotypes by extracting and reviewing standardised patient-level data, supporting clinical reasoning for phenotype development and evaluation. It utilises the OMOP CDM to standardise data abstraction and representation, enabling scalable and interpretable analysis across different data sources for conditions.

I am a clinical epidemiologist working with you. I ran the KEEPER package on CPRD GOLD, a real-world data (RWD) set of longitudinal routinely-collected electronic health records (EHR data) from UK primary care practices. It is standardised in the OMOP CDM format.

The KEEPER profile was generated using a Type 1 Diabetes cohort from the DARWIN EU RWE Catalogue phenotype, informed by DynaMed definition. I received a .JSON file from an investigator who uses this phenotype in her research.

The cohort definition looks like this...

Cohort Entry Events

People with continuous observation of 365 days before event may enter the cohort when observing any of the following:

1. condition occurrences of 'DM1_KA' (a concept set containing codes for Type 1 Diabetes and Ketoacidosis)
2. observations of 'DM1_KA'.
3. procedure occurrences of 'DM1_KA'.

Limit cohort entry events to the earliest event per person.

Inclusion Criteria

1. Previous diagnosis or observation

Entry events with at most 0 of the following criteria:

1. having at least 1 condition occurrence of 'DM1_Prev' (a concept set containing Type 1 Diabetes codes), starting anytime prior to cohort entry start date.
2. having at least 1 condition occurrence of 'DM2_Prev' (a concept set containing Type 2 Diabetes codes), starting anytime prior to cohort entry start date.
3. having at least 1 observation of 'DM1_Prev', starting anytime prior to cohort entry start date.
4. having at least 1 observation of 'DM2_Prev', starting anytime prior to cohort entry start date.
5. having at least 1 procedure occurrence of 'DM1_Prev', starting anytime prior to cohort entry start date.

6. having at least 1 procedure occurrence of 'DM2_Prev', starting anytime prior to cohort entry start date.

2. Previous antidiabetic

Entry events with at most 0 of the following criteria:

1. having at least 1 drug exposure of 'Antidiabetics', starting anytime prior to cohort entry start date.

3. non-insulin antidiabetic after 1y

Entry events with at most 0 of the following criteria:

1. having at least 1 drug era of 'Antidiabetics not Insulin', starting anytime up to 365 days after cohort entry start date.

Cohort Exit

The person exits the cohort at the end of continuous observation. The person exits the cohort when encountering any of the following events:

1. death of any form.

Cohort Eras

Entry events will be combined into cohort eras if they are within 0 days of each other.

I set the package parameter to be sampleSize = 100, which means it should have yielded a random sample of 100 patients. In reality, it yielded 83 unique patients.

Your task is to review each row and use your clinical judgment to assess if the conceptual elements presented represent a “Case”, “Probable” or “No Case” label. I want a label for each row in the spreadsheet (starting after the header row). This is only a phenotype evaluation exercise. You may comment on issues but you are not developing the phenotype. We are evaluating if KEEPER profiles can give enough information to ascribe a clinical label.

Here is logic chart to understand the thinking...

Start

└─ Is EXCL (Alternative Diagnosis) = Yes?

└─ └─ Classify: No case

└─ Else

└─ └─ SYMPTOM = Yes AND TX = Yes → Classify: Case

└─ └─ SYMPTOM = Yes AND Complications = Yes → Classify: Case

└─ └─ SYMPTOM = Yes AND TX = No → Classify: Probable (symptom-only)

└─ └─ SYMPTOM = No AND TX = Yes → Classify: Probable (treatment-only)

└─ └─ SYMPTOM = No AND TX = No → Classify: No case

└─ └─ TX = Yes AND Complications = Yes → Classify: Case

At your earliest convenience, please complete a review of the cases row by row and ascribe a clinical label (Case, Probable, No Case) based on the logic above. You can add this in a new column.

File: Type1_Diabetes_Mellitus.docx
keeper_output_hds3.csv

Deployment

I will use the University of Oxford infrastructure that is configured through Microsoft 365 using single-sign on services for deploying identity & access management (IAM) protocols. My account is provisioned from the University and maintained by University IT. My access is subject to the University's [Policy for using Generative AI in Research](#).

As of September 2025, the University of Oxford holds educational licensing with OpenAI's ChatGPT Edu, Microsoft 365 Copilot, and Google Gemini. Each environment upholds the University's existing IT protocols inclusive of security and access provisions inclusive of other ITS infrastructure, such as the same protocols to configure university servers and analytical layers for utilising data.

The University of Oxford AI Competency Centre maintains resources related to the licensing and infrastructure deployment of these models. Using these central licences offers additional enterprise-level protection for generative AI tools.